In this page, I investigate some typical finite groups.
Specifically, investigate:
Abelian and Abelization
Commutator and Derived subgroup
Upper-Central Series and Solvablity and Composition Series
Center and Conjugacy classes.
Explicit all subgroup

The general Properties of S_n, D_n, C_n will be covered in other page.

$\mathbb{Z}_2 \times S_3$, Conjugacy Classes.

Since $|G : G_a| = |C_a|$, find the Centralizer of $(0, (12))$.

Since $(1,2)$ commutes only id and itself, and $\mathbb{Z}_2$ is abelian,

$$C_G(0,(1,2)) = \{(0,(12)), (0, id), (1,(1,2)), (1, id)\}.$$

Therefore, the conjugacy classes of $(0,(12))$ has $2 \cdot 6/4 = 3$ elements.

Since $\mathbb{Z}_2$ is abelian (or 0 is identity), the first term never change, and since the conjugate cannot change the cycle structure, the conjugacy classes of $(0, (12))$ is

$$\{(0,(12)), (0,(13)), (0,(23))\},$$

and similarly, the conjugacy classes of $(1,(12))$ is

$$\{(1,(12)), (1,(13)), (1,(23))\},$$

" $(0,(123))$ is

$$\{(0,(123)), (0,(132))\}$$

" $(1,(123))$ is

$$\{(1,(123)), (1,(132))\}$$

and $(0, id)$ and $(1, id)$ have conjugacy classes only containing itself.

$\lceil S_3 \times S_3$, Conjugacy classes.

Let $(1\ 2) \times (1\ 2) \in S_3 \times S_3$, $(1\ 2)$ Commute only id or itself, thus

$$C_G((1\ 2)) = \{(id, id), (id, (1\ 2)), ((1\ 2), id), ((1\ 2), (1\ 2))\},$$

So, the conjugacy classes of $(1\ 2) \times (1\ 2)$ has 9 elements

「S_4.

$S_4 =$	$\{ id,$
1,1,2	$(1\ 2), (1\ 3), (1\ 4), (2\ 3), (2\ 4), (3\ 4),$
2,2	$(1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3),$
1,3	$(1\ 2\ 3), (1\ 3\ 2), (1\ 3\ 4), (1\ 4\ 3), (1\ 2\ 4), (1\ 4\ 2), (2\ 3\ 4), (2\ 4\ 3),$
4	$(1\ 2\ 3\ 4), (1\ 2\ 4\ 3), (1\ 3\ 2\ 4), (1\ 3\ 4\ 2), (1\ 4\ 2\ 3), (1\ 4\ 3\ 2)\}$
Partition	

A_4. Conjugacy classes.

$A_4 = \{id, (12)(34), (13)(24), (14)(23),$
$(123), (132), (124), (142), (134), (143),$
$(234), (243)\}$

Since the conjugate cannot change the cycle type, the conjugacy class of each element in A_4 are contained the set of same cycle type. But we don't know it exactly equals to the set, let's calculate.

$$(123)(12)(34)(123)^{-1} = (23)(14)$$

$$(124)(12)(34)(124)^{-1} = (24)(31)$$

therefore, the conjugacy classes of $(12)(34)$ equals the set of $(2,2)$ cycles.

$$(132)(123)(132)^{-1} = (312) = (123) \quad (\text{In actually, } (123)^{-1} = (132))$$

$$(124)(123)(124)^{-1} = (243)$$

$$(142)(123)(142)^{-1} = (413) = (134)$$
$$(134)(123)(134)^{-1} = (324) = (243)$$
$$(143)(123)(143)^{-1} = (421) = (142)$$
$$(234)(123)(234)^{-1} = (134)$$
$$(243)(123)(243)^{-1} = (142)$$
$$(12)(34)(123)((12)(34))^{-1} = (214)$$
$$(13)(24)(123)(13)(24) = (341) = (134).$$
$$(14)(23)(123)(14)(23) = (432) = (243)$$

therefore, the conjugacy classes of (123) has just four elements.

Generally, since $|G : G_a| = C_a$ and $G_a = C_G(a)$ in the conjugate action, therefore number of elements of the orbit is $|G|/|C_G(a)|$.

Since $C_{A_4}((123)) = \{id, (123), (132)\}$,

the orbit of (123) has $12/3 = 4$ elements.

「A_5. The Smallest group of non-abelian Simple group.

「Prop. Group of order 60 having more than one Sylow 5-subgroup is simple.
Proof. Using Contradiction. Suppose that there is a normal subgroup $1 < H \trianglelefteq G$.
By the Sylow Theorem, $n_5 \equiv 1 \bmod 5$ and $n_5 \mid 12$, so $n_5 \in \{1, 6\}$.
By the hypothesis, n_5 must be 6. Let $P \in Syl_5(G)$ be a Sylow 5-subgroup.
Since $n_5 = 6$, that is, $|\{gPg^{-1} \mid g \in G\}| = |G : N_G(P)| = 6$, therefore $|N_G(P)| = 10$.
Assume that $5 \mid |H|$. Then, H contains a Sylow 5-subgroup Q of G,
because $60 = 2^2 \cdot 3 \cdot 5$, and since H is Normal, $gQg^{-1} \subseteq H$ for all $g \in G$,
therefore H contains all Sylow 5-subgroups of G.
But, since for any $P_1, P_2 \in Syl_5(G)$, $P_1 \neq P_2 \Rightarrow P_1 \cap P_2 = \{e\}$,
therefore $|H| \geq 5 \cdot 6 - 6 \cdot 1 + 1 = 25$, so the Lagrange's Theorem forces $|H| = 30$.
But, by the Sylow Theorem, $\# Syl_3(H) \in \{1, 10\}$. If $\# Syl_3(H) = 1$,
then $A \in Syl_3(H)$ must be a normal subgroup of H, then PA is a group of
order $|PA| = |P| \cdot |A| / |P \cap A| = 15/1 = 15$, so $|H : PA| = 30/15 = 2$, $PA \trianglelefteq H$ is normal.
But, order of 5 subgroup in group of 15 is characteristic, because $n_5 \mid 3 \Rightarrow n_5 = 1$,
So P charateristic to PA, and $PA \trianglelefteq H$, so $P \trianglelefteq H$. It contradicts.
Thus, $\# Syl_3(H) = 10$, but every non-identity element of Sylow 3-subgroup has
order 3, and every non-identity element of Sylow 5-subgroup has order 5, so

$$|H| \geq 6 \cdot 4 + 10 \cdot 2 = 24 + 20 = 44$$

But it cannot be happen, since $|H| = 30$. So, $|H|$ is not divided by 5.
Now, $|H| \in \{2, 4, 6, 12\}$. If $|H| = 6$, then by the Sylow theorem,
there is $H_1 < H$ such that $H_1 \in Syl_3(H)$. If $|H| = 12$, then for $n_3 = \# Syl_3(H)$,
$n_3 = 1$ or $n_3 = 4$. But, $n_3 = 4 \Rightarrow |H| \geq n_3 \cdot (3-1) + 1 + n_2 \cdot (4-1) = 9 + 3 \cdot n_2$
$\Rightarrow n_2 = 1$
So there is $H_2 < H$ such that H_2 char H. Anyway, H_1, H_2 char H,
So $H_1, H_2 \trianglelefteq G$. Now, $|G/H_1|, |G/H_2| \in \{15, 10\}$. So $|G/H_i|$ $i = 1, 2$. has
a normal subgroup $K/H_i \trianglelefteq G/H_i$. Now, by the Corresponding isomorphism theorem,
$K \trianglelefteq G$ and $5 \mid |K|$, it contradicts by the above argument.

「Klein 4-group.

Let $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ be the Klein 4-group.

Then, a map $\varphi : G \to S_4$:

$(0,0) \mapsto id$, $(1,0) \mapsto (1\,2)(3\,4)$, $(0,1) \mapsto (1\,3)(2\,4)$, $(1,1) \mapsto (1\,4)(2\,3)$

is Embedding